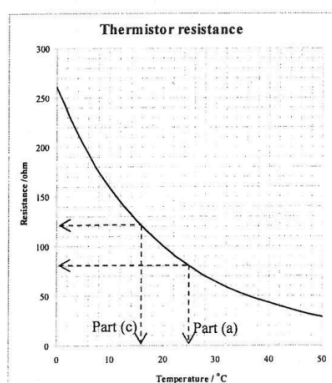


3.	(a)	(i)	$P = F v$	1M	
			$= 8000 \text{ N} \times 2 \text{ m s}^{-1}$	1A	<u>2</u>
			$= 16 \text{ kW}$		
	(b)	(i)	$P_{\text{loss}} = 20 \text{ kW} - 16 \text{ kW} = 4 \text{ kW}$	1A	<u>1</u>
			$P = 4 \text{ kW} + (8000 - 7000 \text{ N}) \times 2 \text{ m s}^{-1}$	1M	
			$= 6000 \text{ W} = 6 \text{ kW}$	1A	<u>2</u>
	(b)	(ii)	Output power required from motor is smaller.	1A	
			<u>Or</u> Force exerted by motor is smaller.		<u>1</u>
	(b)	(iii)	No, the lift system could not work / the lift will fall	1A	
			as slipping would occur / the cable cannot be fixed on the drum / the drum cannot exert a force on the cable.	1A	<u>2</u>

10. (a) (i) $80\ \Omega$ 

$$(ii) \quad V_{AB} = \frac{120}{(80 + 120)} \times 12 = 7.2\text{ V}$$

- (b) As R_v and $120\ \Omega$ resistor are in parallel, R_{eq} across AB is smaller than $120\ \Omega$, therefore voltage shared across AB is reduced / smaller than expected.

Use a voltmeter with resistance much larger than the resistance in that part of the circuit. (e.g. $10\text{ M}\Omega$ in some digital voltmeter)

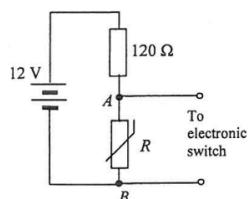
$$(c) (i) \quad V_{AB} = \frac{120}{(R + 120)} \times 12 = 6.0\text{ V}$$

$$R = 120\ \Omega$$

corresponds to temperature at 16°C .

58

10. (c) (ii) Correct circuit (i.e. interchange thermistor R and $120\ \Omega$ resistor). As the temperature drops, the thermistor resistance increases. When the resistance increases to a value such that $V_{AB} = 6.0\text{ V}$ or above, the electronic switch is on and it turns on the heating device.



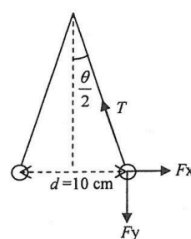
$$11. (a) \quad T \cos \frac{\theta}{2} = F_y = mg$$

$$T \sin \frac{\theta}{2} = F_x = \frac{Q^2}{4\pi \epsilon_0 d^2}$$

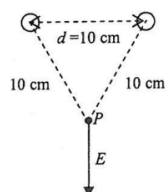
$$\tan \frac{\theta}{2} = \frac{Q^2}{4\pi \epsilon_0 d^2} \left(\frac{1}{mg} \right)$$

$$= 9 \times 10^9 \times \frac{(3.1 \times 10^{-9})^2}{0.1^2} \times \frac{1}{(1.0 \times 10^{-5})(9.81)}$$

$$\frac{\theta}{2} = 5.0^\circ \text{ i.e. } \theta = 10.1^\circ$$



- (b) (i)



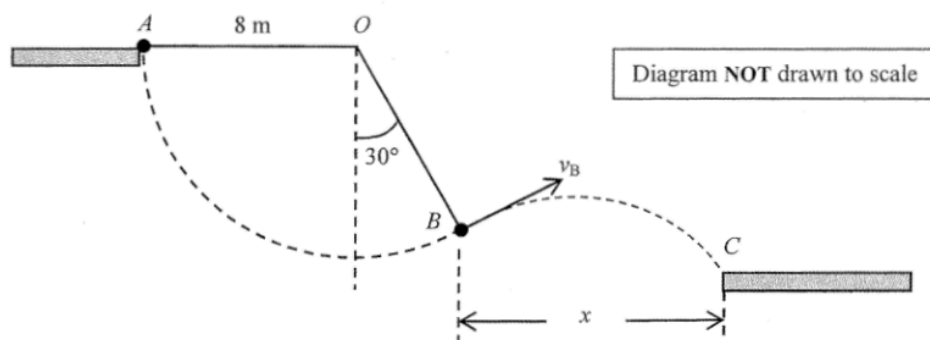
$$(ii) \quad \text{Potential at } P = \frac{Q}{4\pi \epsilon_0 d} + \frac{Q}{4\pi \epsilon_0 d} = \frac{2Q}{4\pi \epsilon_0 d}$$

$$= (9 \times 10^9) \frac{2 \times 3.1 \times 10^{-9}}{0.1}$$

$$= 558\text{ V}$$

- (iii) Separation d decreases.

3. (a)



v_B correctly drawn with label (roughly perpendicular to OB).

$$\frac{1}{2}mv^2 = mgh$$

$$v_B^2 = 2gh = 2 \times 9.81 \times (8 \cos 30^\circ)$$

$$v_B = 11.7 \text{ m s}^{-1} \text{ (11.65896) (or } 11.77 \text{ m s}^{-1} \text{ for } g = 10 \text{ m s}^{-2})$$

(b) (i) $x = v_x t = 11.7 \cos 30^\circ \times 1.25$ [$v_x = v_B \cos 30^\circ$]
 $= 12.6 \text{ m (12.62119)}$
 (or 12.7 to 12.8 m for $g = 10 \text{ m s}^{-2}$)

(ii)

$$y = ut - \frac{1}{2}gt^2$$

$$u = v_y = v_B \sin 30^\circ = 5.83 \text{ m s}^{-1}$$

$$y = v_y(1.25) - \frac{1}{2}(9.81)(1.25)^2$$

$$y = -0.38 \text{ m (-0.414 to -0.352 m)}$$

$$\text{(or } -0.455 \text{ to } -0.4375 \text{ m for } g = 10 \text{ m s}^{-2})$$

Platform C is 0.38 m below B.

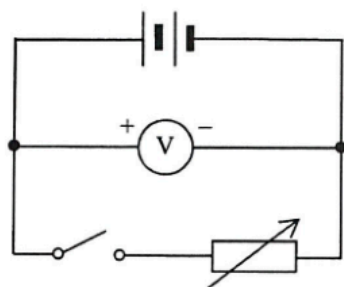
(c) Total mechanical energy is the same / unchanged.

1A
1M
1A
3
1M
1A
2
1M
1M
1A
3
1A
1

7. (a) (i) $R = 10\text{ k}\Omega$ (circuit I) $V = \frac{\left(\frac{1}{10\text{ k}\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}}{10\text{ k}\Omega + \left(\frac{1}{10\text{ k}\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}} \times 6\text{ V}$ $= 2\text{ V}$ $R = 100\text{ }\Omega$ (circuit II) $V = \frac{\left(\frac{1}{100\text{ }\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}}{100\text{ }\Omega + \left(\frac{1}{100\text{ }\Omega} + \frac{1}{10\text{ k}\Omega}\right)^{-1}} \times 6\text{ V}$ $= 2.985\text{ V}$	1M 1A 1A	1M for appropriate method in calculating voltage Note: $100\text{ }\Omega$ and $10\text{ k}\Omega$ in parallel $\approx 99.0099\text{ }\Omega$ Accept stating that V slightly $< 3\text{ V}$
	3	
(ii) Resistance of circuit / that part of circuit would be lowered / altered significantly when introducing the voltmeter (i.e. loading effect). Or The resistance of the voltmeter is comparable to the resistance of resistor R.	1A 1A	
Resistance of voltmeter should be much higher than the resistance of the part of the circuit under study.	1A	
	2	
(b) (i) V_m does NOT give the true voltage for the resistor. $R_m = R_A + R$	1A 1A	
	2	
(ii) For circuit III $R_m = R + R_A = 10 + 1 = 11\text{ }\Omega$ percentage error $= \frac{1\text{ }\Omega}{10\text{ }\Omega} \times 100\%$ $= 10\%$	1M 1A	
	2	

3. (a) (i) (1) $\frac{1}{2}mv^2 = mgh$ $v^2 = 2(9.81)(12)$ $v = 15.344054 \text{ m s}^{-1} \approx 15.3 \text{ m s}^{-1}$ $(v = 15.491933 \text{ m s}^{-1} \approx 15.5 \text{ m s}^{-1} \text{ for } g = 10 \text{ m s}^{-2})$	1M 1A 2	Accept: (15.0 ~ 15.5) m s⁻¹
(2) $s = \frac{1}{2}gt^2$ $12 = \frac{1}{2}(9.81)t^2$ $t = 1.564124 \text{ s} \approx 1.56 \text{ s}$ $(t = 1.5491933 \text{ s} \approx 1.55 \text{ s} \text{ for } g = 10 \text{ m s}^{-2})$	1M 1A 2	Accept: (1.50 ~ 1.60) s
(ii) $F - mg = \frac{mv - mu}{t}$ $F = \frac{70 \times (0 - (-15.3))}{0.3} + 70 \times 9.81$ $= 4266.9793 \text{ N} \approx 4270 \text{ N}$ $(F = 4314.7845 \text{ N} \approx 4310 \text{ N} \text{ for } g = 10 \text{ m s}^{-2})$	1M+1M 1A 3	Accept: (4180 ~ 4320) N
(iii) Elastic potential energy	1A 1	
(b) (i) (Velocity is too high, hence the force for deceleration is very large.) - The life net may be torn. - The falling person may be injured. - The firemen may not be able to hold the life net tight.	1A Any ONE 1	
(ii) There exists a horizontal velocity when a person jumps and the horizontal displacement is very difficult to estimate as it depends on the time of fall, which is relatively long.	1A 1A 2	

7. (a)



Close the switch and record corresponding V and R readings
Adjust the resistance R to lower/other value(s) and repeat the experiment

Precaution:

- First set the variable resistor to its maximum / a large value
 - Open the switch after each measurement
 - Any reasonable answer
- } Any
ONE

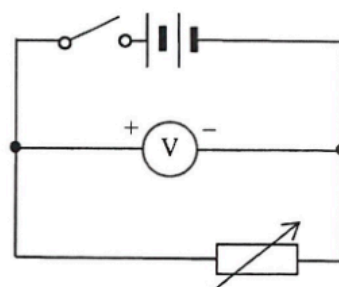
- (b) Terminal voltage V delivered increases with increasing (loading) resistance R (or graphical representation)

$$V = \xi \frac{R}{R+r} \quad \text{OR} \quad V = \xi - \frac{\xi}{R+r} r$$

1A
1A

Correct circuit with correct symbol
Correct polarity

Alternative circuit

1A
1A

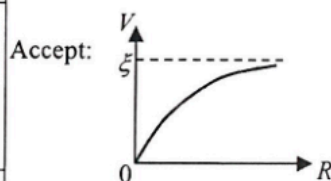
1A

5

1A

1A

2

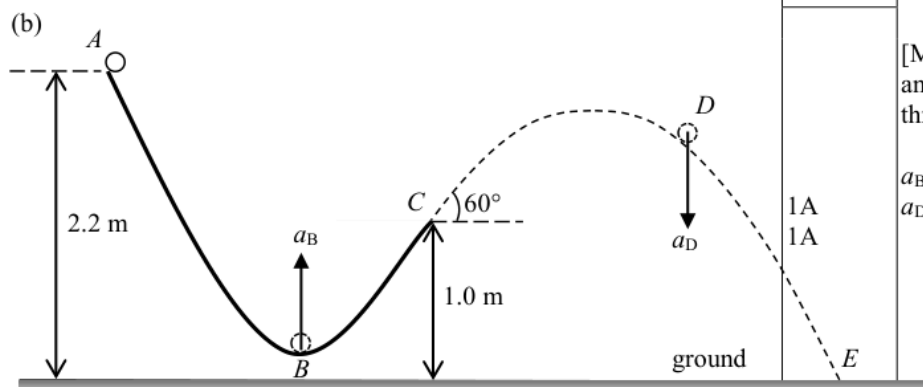


4. (a) $v_B > v_C > v_D > v_A$

1A

[MR: Accept: $v_A < v_D < v_C < v_B$
Accept: B, C, D, A
NOT accept: A, D, C, B]

1

1A
1A

[MR: Accept the arrows drawn at anywhere on the vertical lines passing through the object at positions B & D]

a_B correctly marked
 a_D correctly marked

- (c) (i) Gravitational potential energy is changed to kinetic energy from A to B, and some kinetic energy is changed back to gravitational potential energy from B to C.

1A

2 [MR: 2 marks : All correct and complete: PE → KE & some KE → PE
1 mark :

1A

(i) PE → KE → PE or
(ii) Correctly describe only between 2 points (A → B / B → C / A → C)]

2

(ii) $\frac{1}{2}mv^2 = mgh$

$$v^2 = 2(9.81)(2.2 - 1.0)$$

$$v = 4.852216 \text{ m s}^{-1} \approx 4.85 \text{ m s}^{-1}$$

1M

1A

[MR: h must be correct for 1M
Method using $v^2 - u^2 = 2as$ etc.
is NOT accepted]

For $g = 10 \text{ m s}^{-2}$,

$$v = 4.89898 \text{ m s}^{-1} \approx 4.90 \text{ m s}^{-1}$$

Accept 4.85 m s^{-1} to 4.90 m s^{-1}

2

- (iii) Horizontal speed (at C) = $4.85 \times \cos 60^\circ$
= $2.426108 \text{ m s}^{-1} \approx 2.43 \text{ m s}^{-1}$

1M

e.c.f. from (c)(ii)

For $g = 10 \text{ m s}^{-2}$,

Horizontal speed

$$= 2.44949 \text{ m s}^{-1} \approx 2.45 \text{ m s}^{-1}$$

1M

1A

Vertical speed $\approx 4.24 \text{ m s}^{-1}$

and $t = 1.041033 \text{ s} \approx 1.04 \text{ s}$

Accept 1.04 s to 1.10 s

Or

$$\text{Vertical speed (at C)} = 4.85 \times \sin 60^\circ$$

$$= 4.2002 \text{ m s}^{-1} \approx 4.20 \text{ m s}^{-1}$$

$$-1.0 = 4.20 \times t + \frac{1}{2}(-9.81)t^2$$

$$t = 1.051066 \text{ s} \approx 1.05 \text{ s}$$

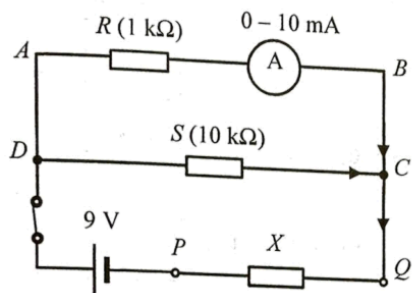
[MR:

1st M for correct expression for initial horizontal /vertical speed

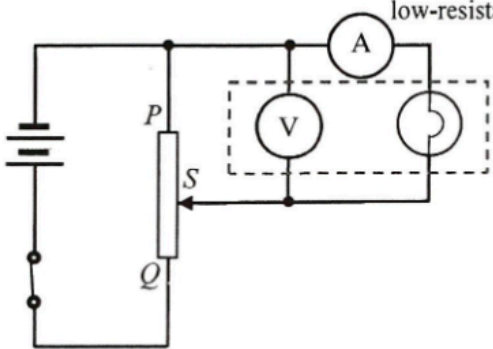
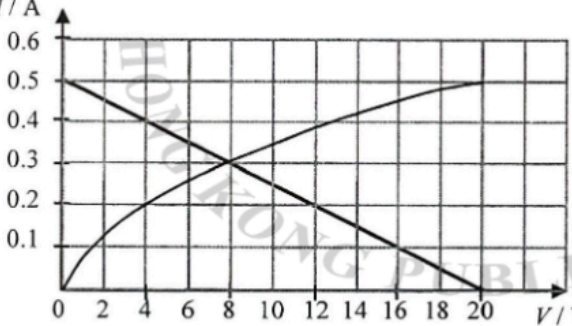
2nd M for correct method (equation) to find t by considering horizontal motion/ vertical motion

3

63

	Marks	Remarks
8. (a) (i) $V_{AB} = (8.5 \times 10^{-3} \text{ A}) \times (1 \text{ k}\Omega)$ $= 8.5 \text{ V}$	1M 1A	
(ii) Current through $S = \frac{8.5 \text{ V}}{10 \text{ k}\Omega}$ $= 8.50 \times 10^{-4} \text{ A} = 0.85 \text{ mA}$	2 1M 1A	As AB and CD are connected in parallel, current through S $= (8.5 \times 10^{-3} \text{ A}) \times \frac{1 \text{ k}\Omega}{10 \text{ k}\Omega}$
(iii) 	2A 2	
(iv) p.d. across $X = 9 - V_{AB} = 9 - 8.5 = 0.5 \text{ V}$ Current through $X = (8.5 \times 10^{-3}) + (8.5 \times 10^{-4})$ $= 9.35 \times 10^{-3} \text{ A}$ Resistance of $X = \frac{0.5 \text{ V}}{9.35 \times 10^{-3} \text{ A}}$ $= 53.476 \Omega \approx 53.5 \Omega$	1A 1M 1A	
(b) To prevent the ammeter from overloading (when PQ is shorted or the resistance to be measured is too small)	3 1A 1	

4. (a)	$\frac{1}{2}mv^2 = mgh$ $= (50)(9.81)(1.5)$ $= 735.75 \text{ J} \approx 736 \text{ J}$	1M	For $g = 10 \text{ m s}^{-2}$, K.E. = 750 J
		1A	
		2	
		2A	
		2	
		1M	
(b) (i)	Kinetic energy and potential energy (of the athlete) change to elastic potential energy (of the trampoline).	1M	For $g = 10 \text{ m s}^{-2}$, $\bar{F} = 2375 \text{ N}$
		1A	
		2	
		2A	
		2	
		1M	
(ii)	$\bar{F}d = mgh + mgd \quad [\text{Or } \bar{F}d = \frac{1}{2}mv^2 + mgd]$ $\bar{F} = \frac{50(9.81)(1.5+0.40)}{0.40}$ $= 1839.375 + 490.5$ $= 2329.875 \text{ N} \approx 2330 \text{ N}$	1M	
		1A	
		2	
		2A	
		2	
		1M	
Or	$\bar{F} = \frac{1}{0.40} [735.75 + (50)(9.81)(0.40)]$ $\approx 2330 \text{ N}$	1M	
		1A	
		2	
		2A	
		2	
		1M	

Solution	Marks	Remarks
8. (a) (i) 	low-resistance ammeter 1A	
(ii) Brightness increases.	1A	
(b) (i) Resistance = $\frac{V}{I} = \frac{20}{0.5}$ = 40 Ω	1M 1A	
(ii) As the applied voltage V increases, current I / electrical power increases and this raises the filament's temperature, thus the resistance R of the filament / bulb increases. Or the electrons collide with the increasingly vibrating atoms / lattice ions (of the filament) which impede their flow, i.e. R increases.	1A 1A 1A	
(c) (i)  Current = 0.3 A (at $V = 8$ V)	1M 1A	Correct straight line intersects the curve at (8, 0.3)
(ii) Power consumed = IV = (0.3)(8) = 2.4 W	1M 1A	e.c.f. from (c)(i)

7.	(a)	$R = \rho \frac{l}{A} \Rightarrow 0.50 = \rho \frac{0.20}{8.0 \times 10^{-7}}$ $\rho = 2.0 \times 10^{-6} (\Omega \text{ m})$	1M	
			1A	
				2
	(b)	(i) Terminal 3	1A	
				1
		(ii) When switch S is connected to terminal 2, the equivalent resistance across MN		
			1M	
			1A	
				2
		(iii) (p.d. across R_2 is) greater than (p.d. across R_1)	1A	
				1
		(iv) zero as no p.d. across CF / no current in the coil.	1A	
			1A	
				2

11. (a)	As the foil becomes longer / l increases while its cross-sectional area becomes smaller / A decreases due to decrease in width w , its resistance would increase according to $R = \frac{\rho l}{A}$.	1A	
		1A	
		2	
(b) (i)	$V_{in} = I_1(R_1 + R_2)$	1A	
		1	
(ii)	Before: $\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{500}{470 + 500} - \frac{470}{470 + 470} = 0.0154639$ After: $\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$ $= \frac{501}{470 + 501} - \frac{470}{470 + 470} = 0.0159629$ Thus $\frac{V_{out}}{V_{in}}$ increases $3.22687\% \approx 3.23\%$.	1M	
		1A	Accept $\pm(3.2 \sim 3.23)\%$ as the change
		2	
(iii)	(I) crack ① (II) along AB	1A 1A	
		2	

2.	(a)	$s = ut + \frac{1}{2}at^2$			
		$0 = 5t + \frac{1}{2}(-9.81)t^2$			
		$t = 1.0194 \text{ s} \approx 1.02 \text{ s}$ or 0 s (rejected)	1M		Or $v = u + at,$ $0 = 5 + (-9.81)(t/2)$ $t = 1.02 \text{ s}$
	(b)	speed of the bead at its maximum height = 2 m s^{-1}	1A		
		$\text{K.E.} = \frac{1}{2} \times 0.01 \times 2^2$		2	
		$= 0.02 \text{ J}$	1M		Accept: $t = 1.00 \sim 1.02 \text{ s}$
	(c)	The bead and the trolley have the same horizontal velocity and thus travel the same horizontal distance (within the same time interval t).	1A		
				2	
			1A		
				2	

7.	(a)	To protect the ammeter / limiting the current in the circuit / ensuring the circuit does not over-heat.	1A		
				1	
			1M		
	(b)	$R = \frac{V}{I} = \frac{3.5}{0.7}$ $= 5.0 \Omega (> 4.7 \Omega)$ The extra resistance (0.3Ω) may be due to the resistance of the ammeter / contact resistances / internal resistance of the power supply.	1A		
			1A		
				3	
	(c)	$I = 0.25 \text{ A}$ at $l = 30 \text{ cm}$ $\Delta R = R_{AC} = \frac{3.5}{0.25} - 5.0$ $= 9.0 \Omega$			Accept: $I = 0.24 \text{ A} \sim 0.26 \text{ A}$
			1M		e.c.f. from (b)
			1A		Accept: $\Delta R = 8.5 \Omega \sim 9.6 \Omega$
				2	

$$= 1.44 \times 10^{14} \text{ J} \approx 1.4 \times 10^8 \text{ (MJ)}$$

$$(ii) \quad E_{out} = (680 \times 10^6) \times (24 \times 3600) \\ = 5.875 \times 10^{13} \text{ J} \approx 5.9 \times 10^7 \text{ (MJ)}$$

$$(iii) \quad \text{Efficiency} = \frac{E_{out}}{E_{in}} \times 100\% \\ = \frac{5.88 \times 10^{13}}{1.44 \times 10^{14}} \times 100\% \\ = 40.8\% \approx 41\%$$

$$(b) \quad (i) \quad P = VI \\ 680 \text{ MW} = 400 \text{ kV} \times I \\ I = 1700 \text{ A} = 1.7 \text{ kA}$$

$$(ii) \quad \text{Power loss} = I^2 R \\ = (1.7 \times 10^3)^2 \times (2.90 \times 10^{-8} \times 160 \times 10^3) \\ = 13410 \text{ W} \approx 13.4 \text{ (kW)}$$

(iii) Only a.c. voltage can be stepped up (or stepped down) using a transformer, which is cost effective. High voltage transmission has low transmission current and so reduces power loss significantly.

1M	2
1A	
2	
1M	
1A	
2	
1M	
1A	
2	
1A	
1A	
2	

e.c.f. from (a)(i)(ii)

Accept: 41% ~ 42%

e.c.f. from (b)(i)